

ELPD estimation

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ELPD estimation for joint model

Let us denote $\theta = (B, \Sigma, \phi)^\top$ as the collection of model parameters. The joint predictive density of y^* (of dimension $u \times q$) given y (of dimension $n \times q$) is

$$\begin{aligned}\pi(y^* | y) &= \int \int \int \pi(y^* | w^*) \pi(w^* | w, \theta) \pi(w, \theta | y) dw^* dw d\theta \\ &= \int \int \int \prod_{i=1}^u \prod_{j=1}^q \pi_j(y_j(s_i^*) | w_j(s_i^*)) \pi(w^* | w, \theta) \pi(w, \theta | y) dw^* dw d\theta.\end{aligned}\tag{1}$$

The marginal ELPD of Y^* given the observed data Y is written as:

$$\text{ELPD} = \log[\pi(y^* | y)]\tag{2}$$

The estimated ELPD is therefore given by,

$$\widehat{\text{ELPD}}_{\text{joint}} = \log \left[\frac{1}{T} \sum_{t=1}^T \prod_{i=1}^u \prod_{j=1}^q \pi_j(y_j(s_i^*) | w_j^{*(t)}(s_i^*)) \right].\tag{3}$$

where $w_j^{*(t)}$ are the t -th posterior draw from the “joint” conditional distribution of W^* given (W, B, Σ, ϕ) which is given by

$$W^* | W, B, \Sigma, \phi \sim \mathcal{MN}_{u,q} \left(X^* B - (Q_{\phi, (m)}^{(u,u)})^{-1} Q_{\phi, (m)}^{(u,n)} (W - XB), (Q_{\phi, (m)}^{(u,u)})^{-1}, \Sigma \right),\tag{4}$$

ELPD estimation for separate model

The predictive density of y^* (of dimension $u \times q$) given y (of dimension $n \times q$) for the separate model can be decomposed as

$$\pi(y^* | y) = \prod_{j=1}^q \pi_j(y_j^* | y_j)\tag{5}$$

The marginal ELPD of Y^* given the observed data Y is written as:

$$\text{ELPD} = \sum_{j=1}^q \log[\pi(y_j^* | y_j)]\tag{6}$$

The estimated ELPD for the separate model thus be given as

$$\widehat{\text{ELPD}}_{sep} = \sum_{j=1}^q \log \left[\frac{1}{T} \sum_{t=1}^T \prod_{i=1}^u \pi_j(y_j(s_i^*) | w_j^{*(t)}(s_i^*)) \right]. \quad (7)$$

where $w_j^{*(t)}$ are the t -th posterior draw from the “separate” conditional distribution of W_j^* , $j = 1, \dots, q$ given $(W_j, \beta_j, \Sigma_{jj}, \phi_j)$ which is given by

$$W_j^* | W_j, \beta_j, \Sigma_{jj}, \phi_j \sim \mathcal{MN}_{u,1} \left(X^* \beta_j - (Q_{\phi_j, (m)}^{(u,u)})^{-1} Q_{\phi_j, (m)}^{(u,n)} (W_j - X \beta_j), (Q_{\phi_j, (m)}^{(u,u)})^{-1}, \Sigma_{jj} \right), \quad (8)$$

Results:

The estimated ELPDs for joint and separate model is -50.645 and -54.886 respectively.